



# SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



|                                                                                                                   |                              |
|-------------------------------------------------------------------------------------------------------------------|------------------------------|
| LOCATION: Jetty 1                                                                                                 | AREA: B                      |
| PERMIT TO WORK No.: -                                                                                             | TYPE OF SCAFFOLD: cantilever |
| SCAFFOLD DRAWING No: PMS-WA-ARCO-037                                                                              |                              |
| RISK CLASSIFICATION: <input checked="" type="radio"/> High <input type="radio"/> Medium <input type="radio"/> Low |                              |

## REFERENCE STANDARDS

- BS EN 1991-1-4:2005 - Wind Actions
- BS EN 1993-1-1:2005 - Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 - Temporary Works Equipment - Scaffolds

## SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

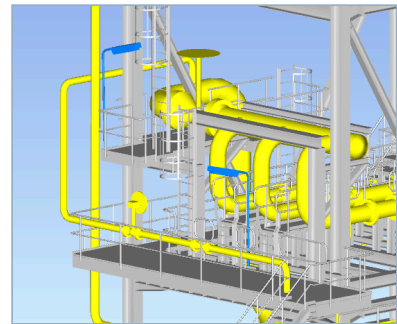
### BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of a cantilever scaffold required for maintenance of in-let piping to A-3302 for equipment tag A-3302 under work order 10086944 at Jetty 1 / B. The scaffold envelope is 7.2m x 3.5m x 3m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, access, and support/tie arrangements as indicated on the approved drawing.

### SAFE WORKING LOAD (SWL): 1.5 kN/m<sup>2</sup>

**Note:** This is a category 1 engineered scaffold. Scaffold shall be inspected and approved for use by assurance engineer before tagging for use.

### SITE / EQUIPMENT PHOTO



| Role        | ID          | Name               | Signature | Date |
|-------------|-------------|--------------------|-----------|------|
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Document No: PMS-WA-ARCO-037

Revision: 0

Document Life: 3 Years

Prepared: 19-Jun-2026

ERMT NO: PIC/PMS/26/2547

|                                                                                   |                  |                        |                               |                                                                                     |
|-----------------------------------------------------------------------------------|------------------|------------------------|-------------------------------|-------------------------------------------------------------------------------------|
|  | Project Name:    | SCAFFOLD DESIGN REPORT |                               |  |
|                                                                                   | Scaffold Type:   | cantilever             | Calculation Sheet             |                                                                                     |
|                                                                                   | Document Number: | PMS-WA-ARCO-037        | Prepared By: Nnamani, Ifeanyi |                                                                                     |

LOAD SUMMARY

TOTAL HORIZONTAL LOAD SUMMARY

| Direction | Wind (kN) | Handrail (kN) | TOTAL (kN) |
|-----------|-----------|---------------|------------|
| X         | 5.36      | 2.1           | 7.46       |
| Z         | 11.64     | 1.44          | 13.08      |

TOTAL VERTICAL LOAD SUMMARY

| Load Case          | Load Description | TOTAL LIVE LOAD, FY (kN) |
|--------------------|------------------|--------------------------|
| LC 2               | LL (Live Load)   | 12.6                     |
| Governing LL Total |                  | 12.6                     |

NET GLOBAL REACTION ON PARENT STRUCTURE

Metric used here: **Net Global Reaction**. This is the true vector sum at the tie/support nodes, preserving FX and FZ signs so opposing directions are allowed to cancel. It is quoted on this page because it represents the net horizontal action transferred to the parent structure. Total Restraint Demand is a separate no-cancellation metric and is tabulated only in Appendix E. Relevant tie nodes: 7, 8, 19, 20, 21, 22, 27.

| Metric                         | Governing LC | Load Description                | Sum FX (kN) | Sum FZ (kN) | Resultant H (kN) |
|--------------------------------|--------------|---------------------------------|-------------|-------------|------------------|
| Governing Net Global Resultant | LC 31        | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLZ | -0.01       | -19.62      | 19.62            |

|                                                                                   |                  |                        |                               |                                                                                     |
|-----------------------------------------------------------------------------------|------------------|------------------------|-------------------------------|-------------------------------------------------------------------------------------|
|  | Project Name:    | SCAFFOLD DESIGN REPORT |                               |  |
|                                                                                   | Scaffold Type:   | cantilever             | Calculation Sheet             |                                                                                     |
|                                                                                   | Document Number: | PMS-WA-ARCO-037        | Prepared By: Nnamani, Ifeanyi |                                                                                     |

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|                                                                                   |                  |                        |                               |                                                                                     |
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|  | Project Name:    | SCAFFOLD DESIGN REPORT |                               |  |
|                                                                                   | Scaffold Type:   | cantilever             | Calculation Sheet             |                                                                                     |
|                                                                                   | Document Number: | PMS-WA-ARCO-037        | Prepared By: Nnamani, Ifeanyi |                                                                                     |

## GENERAL NOTE

### BS EN 12811-1, 4.2.1.2

Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm<sup>2</sup>. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

## Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

## Modelling Philosophy

### BS EN 12811-1, 4.2.1.2

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ( $KFX/KFZ = 13.574 \text{ kN}$ ). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See [Appendix A](#) for model views.

## LOAD CASES

Load diagrams are shown in [Appendix B](#).

### Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m<sup>3</sup>**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

## Live Load on Platforms

### BS EN 12811-1

Platform live loading was applied to the scaffold working platforms as member UDLs in the STAAD model. The platform load intensity is multiplied by the tributary width to obtain the line load applied to platform supporting members. (LC 2).

| Member                                            | Tributary Width (m)       | Service Load (kN/m <sup>2</sup> ) | Member Load (kN/m) |
|---------------------------------------------------|---------------------------|-----------------------------------|--------------------|
| Members 32, 38, 46, 47, 106, 108, 111, 113 (Edge) | $1.200 / 2 = 0.600$       | 1.500                             | <b>0.900</b>       |
| Members 51, 52, 107, 112 (Interior)               | $(0.600 + 0.600) = 1.200$ | 1.500                             | <b>1.800</b>       |

Member Load = Service Load x Tributary Width. Edge members take half the adjacent bay width; interior members take half the bay width from each side.

Governing platform live load intensity used for SWL = **1.5 kN/m<sup>2</sup>**.

Total applied platform live load from STAAD output = **12.6 kN**.

## Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **2.1 kN** (member UDL 0.15 kN/m)

Handrail Load (HLZ) = **1.44 kN** (member UDL 0.15 kN/m)

## Wind Load

### EN 1991-1-4:2005

Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.07928 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load - X Axis (WLX) = **5.36 kN**  
Total Wind Load - Z Axis (WLZ) = **11.64 kN**

### WIND LOAD CALCULATION - EN 1991-1-4:2005

| Reference                                                        | Calculation                                                                                   | Value            | Unit        |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------|-------------|
|                                                                  | Basic wind speed (3-second gust)                                                              |                  |             |
|                                                                  | V3 =                                                                                          | <b>36.0</b>      | m/s         |
|                                                                  | <i>Eurocode <math>V_{b,0}</math> is the characteristic 10-minute mean wind velocity</i>       |                  |             |
| Dredger P. - How to Approximate                                  | Convert 3-second gust to 10-minute mean wind velocity                                         |                  |             |
|                                                                  | V3 / V3600 =                                                                                  | 1.52             |             |
|                                                                  | V600 / V3600 =                                                                                | 1.06             |             |
|                                                                  | V600 = (V600/V3600) x (V3600/V3) x V3 =                                                       | <b>25.11</b>     | m/s         |
|                                                                  | Basic wind speed $V_b = V600 =$                                                               | <b>25.11</b>     | m/s         |
| EN 1991-1-4:2005                                                 | Roughness Length (z0) - Terrain Category 0                                                    | 0.003            | m           |
|                                                                  | Terrain category II (z0,ii)                                                                   | 0.05             |             |
| EN 1991-1-4, Table 4.1                                           | Z <sub>min</sub>                                                                              | 1.0              | m           |
|                                                                  | <b>z - height of structure under consideration</b>                                            | <b>23.0</b>      | m           |
| EN 1991-1-4, Sec 4.3                                             | C <sub>r0</sub> (z) - Roughness factor                                                        | 1.0              |             |
| EN 1991-1-4, Sec 4.3                                             | C <sub>o</sub> (z) - Orography factor                                                         | 1.0              |             |
| EN 1991-1-4, Sec 4.3                                             | Terrain factor $K_r = 0.19 \times [z0/z0,ii]^{0.07}$                                          | 0.156            |             |
| EN 1991-1-4, Table 4.1                                           | Z <sub>min</sub> < z -> Case 1 applies                                                        |                  |             |
|                                                                  | <b>Wind Turbulence</b>                                                                        |                  |             |
|                                                                  | C <sub>r</sub> (z) = K <sub>r</sub> x ln(z/z0)                                                | 1.3957           |             |
|                                                                  | Mean velocity V <sub>m</sub> (z) = C <sub>r</sub> (z) x C <sub>o</sub> (z) x V <sub>b</sub>   | 35.039           | m/s         |
| EN 1991-1-4, Sec 4.4                                             | K1                                                                                            | 1.0              |             |
|                                                                  | Turbulence intensity I <sub>v</sub> (z) = k1 / [C <sub>o</sub> (z) x ln(z/z0)]                | 0.1118           |             |
|                                                                  | <b>Peak Velocity Pressure</b>                                                                 |                  |             |
|                                                                  | Density of air rho                                                                            | 1.25             | kg/m³       |
| EN 1991-1-4, Sec 4.5                                             | q <sub>p</sub> (z) = [1 + 7*I <sub>v</sub> (z)] x 0.5 x rho x V <sub>m</sub> (z) <sup>2</sup> | <b>1367.8401</b> | N/m²        |
| <b>Wind Load = C<sub>f</sub> x q<sub>p</sub> x A<sub>f</sub></b> |                                                                                               |                  |             |
| C <sub>f</sub> =                                                 |                                                                                               | 1.2              |             |
| q <sub>p</sub> =                                                 |                                                                                               | 1.36784          | kN/m²       |
| A <sub>f</sub> =                                                 | Tube outer diameter (48.3 mm)                                                                 | 0.0483           | m           |
| <b>Wind Load UDL =</b>                                           |                                                                                               | <b>0.07928</b>   | <b>kN/m</b> |

## LOAD COMBINATIONS

### BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;  
SLS (characteristic) combinations use gamma = **1.0**.

**DL** = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

| LC No. | Combination                     | Component Loadings & Factors                  |
|--------|---------------------------------|-----------------------------------------------|
| 7      | DL + LL                         | LC1 x 1.0 + LC2 x 1.0                         |
| 8      | DL + HLX                        | LC1 x 1.0 + LC3 x 1.0                         |
| 9      | DL + HLZ                        | LC1 x 1.0 + LC4 x 1.0                         |
| 10     | DL + WLX                        | LC1 x 1.0 + LC5 x 1.0                         |
| 11     | DL + WLZ                        | LC1 x 1.0 + LC6 x 1.0                         |
| 12     | DL + LL + HLX                   | LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0             |
| 13     | DL + LL + HLZ                   | LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0             |
| 14     | DL + LL + WLX                   | LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0             |
| 15     | DL + LL + WLZ                   | LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0             |
| 16     | DL + LL + HLX + WLX             | LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC5 x 1.0 |
| 17     | DL + LL + HLZ + WLZ             | LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC6 x 1.0 |
| 18     | DL + LL + HLX + WLZ             | LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC6 x 1.0 |
| 19     | DL + LL + HLZ + WLX             | LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC5 x 1.0 |
| 20     | 1.5DL                           | LC1 x 1.5                                     |
| 21     | 1.5DL + 1.5LL                   | LC1 x 1.5 + LC2 x 1.5                         |
| 22     | 1.5DL + 1.5HLX                  | LC1 x 1.5 + LC3 x 1.5                         |
| 23     | 1.5DL + 1.5HLZ                  | LC1 x 1.5 + LC4 x 1.5                         |
| 24     | 1.5DL + 1.5WLX                  | LC1 x 1.5 + LC5 x 1.5                         |
| 25     | 1.5DL + 1.5WLZ                  | LC1 x 1.5 + LC6 x 1.5                         |
| 26     | 1.5DL + 1.5LL + 1.5HLX          | LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5             |
| 27     | 1.5DL + 1.5LL + 1.5HLZ          | LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5             |
| 28     | 1.5DL + 1.5LL + 1.5WLX          | LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5             |
| 29     | 1.5DL + 1.5LL + 1.5WLZ          | LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5             |
| 30     | 1.5DL + 1.5LL + 1.5HLX + 1.5WLX | LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC5 x 1.5 |
| 31     | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLZ | LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC6 x 1.5 |
| 32     | 1.5DL + 1.5LL + 1.5HLX + 1.5WLZ | LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC6 x 1.5 |
| 33     | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLX | LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC5 x 1.5 |

## STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

**Maximum Utilization Ratio = 0.579** at Member No. **45**, Load Combination **31**, Clause **EC-6.2.7(5)**. **PASS** - All members adequate; UC ratio below unity.

UC ratio diagram: see [Appendix C](#).

| Member | Section  | Status | Clause       | UC Ratio     | L/C | F <sub>x</sub> (kN) | Type |
|--------|----------|--------|--------------|--------------|-----|---------------------|------|
| 45     | PIP483.6 | PASS   | EC-6.2.7(5)  | <b>0.579</b> | 31  | 2.72                | T    |
| 49     | PIP483.6 | PASS   | EC-6.2.7(5)  | <b>0.560</b> | 32  | 2.76                | T    |
| 63     | PIP483.6 | PASS   | EC-6.2.9.1   | <b>0.456</b> | 30  | 0.29                | C    |
| 105    | PIP483.6 | PASS   | EC-6.3.3-661 | <b>0.441</b> | 32  | 9.72                | C    |
| 112    | PIP483.6 | PASS   | EC-6.2.9.1   | <b>0.395</b> | 31  | 0.94                | T    |
| 108    | PIP483.6 | PASS   | EC-6.3.3-662 | <b>0.380</b> | 31  | 0.51                | C    |
| 106    | PIP483.6 | PASS   | EC-6.2.9.1   | <b>0.348</b> | 31  | 0.93                | T    |
| 64     | PIP483.6 | PASS   | EC-6.2.9.1   | <b>0.334</b> | 31  | 1.71                | T    |

| Member | Section  | Status | Clause       | UC Ratio | L/C | F <sub>x</sub> (kN) | Type |
|--------|----------|--------|--------------|----------|-----|---------------------|------|
| 51     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.331    | 21  | 0.76                | T    |
| 62     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.325    | 31  | 1.54                | C    |
| 104    | PIP483.6 | PASS   | EC-6.3.3-661 | 0.310    | 32  | 6.31                | C    |
| 60     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.307    | 31  | 1.62                | C    |
| 47     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.304    | 30  | 1.04                | T    |
| 61     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.298    | 30  | 1.57                | C    |
| 38     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.294    | 21  | 0.00                | C    |
| 109    | PIP483.6 | PASS   | EC-6.2.9.1   | 0.281    | 31  | 0.12                | T    |
| 110    | PIP483.6 | PASS   | EC-6.2.9.1   | 0.281    | 31  | 0.11                | T    |
| 29     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.278    | 30  | 3.71                | C    |
| 65     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.269    | 30  | 1.59                | T    |
| 115    | PIP483.6 | PASS   | EC-6.2.9.1   | 0.266    | 21  | 0.03                | T    |
| 114    | PIP483.6 | PASS   | EC-6.2.9.1   | 0.264    | 21  | 0.03                | T    |
| 103    | PIP483.6 | PASS   | EC-6.3.3-661 | 0.262    | 32  | 5.08                | C    |
| 113    | PIP483.6 | PASS   | EC-6.2.9.1   | 0.253    | 31  | 2.03                | T    |
| 111    | PIP483.6 | PASS   | EC-6.3.3-662 | 0.252    | 31  | 1.41                | C    |
| 85     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.244    | 31  | 0.26                | C    |
| 70     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.230    | 30  | 0.16                | C    |
| 71     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.230    | 30  | 0.16                | C    |
| 95     | PIP483.6 | PASS   | EC-6.3.3-661 | 0.216    | 32  | 3.87                | C    |
| 59     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.214    | 21  | 1.23                | C    |
| 66     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.190    | 31  | 0.27                | C    |
| 83     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.188    | 30  | 0.62                | C    |
| 84     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.188    | 30  | 0.62                | C    |
| 32     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.185    | 21  | 0.00                | C    |
| 91     | PIP483.6 | PASS   | EC-6.3.3-661 | 0.183    | 32  | 3.02                | C    |
| 41     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.180    | 30  | 4.08                | T    |
| 67     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.175    | 21  | 0.20                | C    |
| 79     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.173    | 31  | 0.93                | C    |
| 48     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.169    | 30  | 6.44                | T    |
| 50     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.169    | 30  | 6.44                | T    |
| 46     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.167    | 30  | 1.04                | T    |
| 3      | PIP483.6 | PASS   | EC-6.3.3-662 | 0.164    | 32  | 5.14                | C    |
| 23     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.163    | 31  | 3.75                | C    |
| 44     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.154    | 31  | 5.91                | T    |
| 80     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.154    | 21  | 0.91                | C    |
| 68     | PIP483.6 | PASS   | EC-6.3.3-662 | 0.146    | 31  | 1.61                | C    |
| 5      | PIP483.6 | PASS   | EC-6.3.3-662 | 0.144    | 32  | 4.21                | C    |
| 74     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.144    | 30  | 0.66                | C    |
| 75     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.144    | 30  | 0.66                | C    |
| 43     | PIP483.6 | PASS   | EC-6.2.9.1   | 0.142    | 31  | 3.93                | T    |
| 107    | PIP483.6 | PASS   | EC-6.2.9.1   | 0.142    | 27  | 0.71                | T    |

Showing highest 50 utilization ratios; 58 lower-utilization members omitted.

## SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **4.831 kN** (Member 44, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 4.831 kN <= 10.0 kN -> **PASS**

Class A right-angle couplers are adequate.

Member force table: see [Appendix F](#)

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

| Coupler Type                 | Resistance                               | Characteristic Value |         |
|------------------------------|------------------------------------------|----------------------|---------|
|                              |                                          | Class A              | Class B |
| Right-angle Coupler (RA)     | Slipping force $F_{s,k}$ (kN)            | 10.0                 | 15.0    |
|                              | Cruciform bending moment $M_{B,k}$ (kNm) | -                    | 0.8     |
|                              | Pull-apart force $F_{p,k}$ (kN)          | 20.0                 | 30.0    |
|                              | Rotational moment $M_{T,k}$ (kNm)        | -                    | 0.13    |
| Swivel Coupler (SW)          | Slipping force $F_{s,k}$ (kN)            | 6.0                  | 9.0     |
|                              | Cruciform bending moment $M_{B,k}$ (kNm) | -                    | 0.8     |
|                              | Pull-apart force $F_{p,k}$ (kN)          | 15.0                 | 22.5    |
|                              | Rotational moment $M_{T,k}$ (kNm)        | -                    | 0.08    |
| Putlog / Single Coupler (PL) | Slipping force $F_{s,k}$ (kN)            | -                    | 3.0     |
|                              | Cruciform bending moment $M_{B,k}$ (kNm) | -                    | 0.6     |

### FRICTIONAL RESISTANCE

Max vertical reaction per support (Max  $f_y$ ) = Resistance x 0.15 /  $C_F$  = **6.787 kN**

Coefficient of friction ( $C_F$ ) = **0.3**

Frictional Resistance (FR) =  $C_F$  x Max  $f_y$  = **2.0361 kN**

Spring Stiffness = FR / 0.15 = **13.574 kN** <- applied as KFX/KFZ in STAAD model

### DEFLECTION CHECK RESULTS

**BS EN 12811-1, Cl 6.3** Serviceability limit state - values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in [Appendix C](#) and [Appendix F](#).

#### Vertical Deflection Check

| Parameter                                                                    | Value    | Allowable                    | Check |
|------------------------------------------------------------------------------|----------|------------------------------|-------|
| Max vertical displacement - LC 16<br>Member 41   Member length L = 2400.0 mm | 2.059 mm | $L/100 = 2400/100 = 24.0$ mm | PASS  |

#### Horizontal Deflection Check

| Parameter                                                                      | Value    | Allowable                   | Check |
|--------------------------------------------------------------------------------|----------|-----------------------------|-------|
| Max horizontal displacement - LC 16<br>Member 29   Member length L = 1700.0 mm | 1.534 mm | $L/200 = 1700/200 = 8.5$ mm | PASS  |

### CONCLUSION

The structural analysis and design of the **cantilever** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.579** at Member 45). Vertical and horizontal deflections are



within their respective allowable limits of  $L/100 = 24.0\text{mm}$  and  $L/200 = 8.5\text{mm}$ . All scaffold tube coupler connections are adequate for the applied axial loads.

## RECOMMENDATIONS

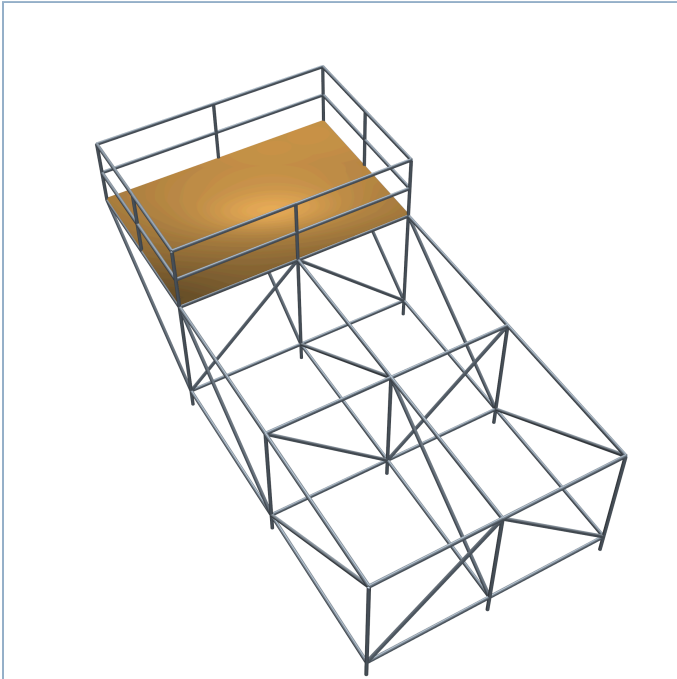
The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

## INSTALLATION NOTES

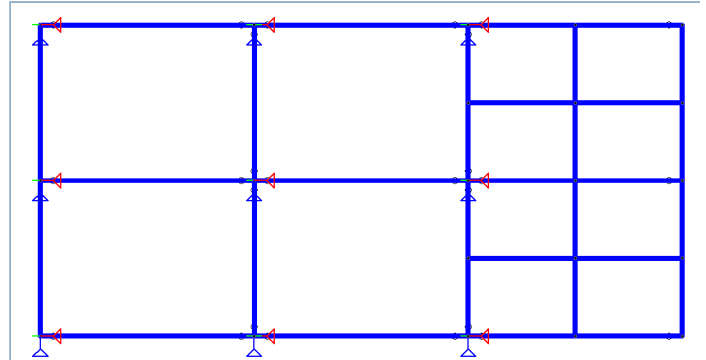
1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the cantilever scaffold configuration matches the approved drawing and is suitable for maintenance of in-let piping to A-3302.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Install and secure cantilever support members, needles, anchors, ties, and back-span restraints exactly as shown on the approved drawing before building the working platform.
8. Build the scaffold frame to the approved envelope of 7.2m x 3.5m x 3.0m, checking level, plumb, and cantilever projection at each lift.
9. Install bracing, ledgers, transoms, platform boards, guardrails, and toe boards progressively, maintaining all specified ties and restraints.
10. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
11. Install mid-level horizontal ledgers and transoms at 2.5m height in accordance with the working drawing.
12. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
13. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
14. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
15. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 0.3m and 2.0m height as shown on the working drawing.
16. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
17. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
18. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.  
End-user or third-party modification of this scaffold structure is strictly prohibited.
19. For dismantling, perform the erection steps in reverse order.

## APPENDIX A - MODEL VIEWS

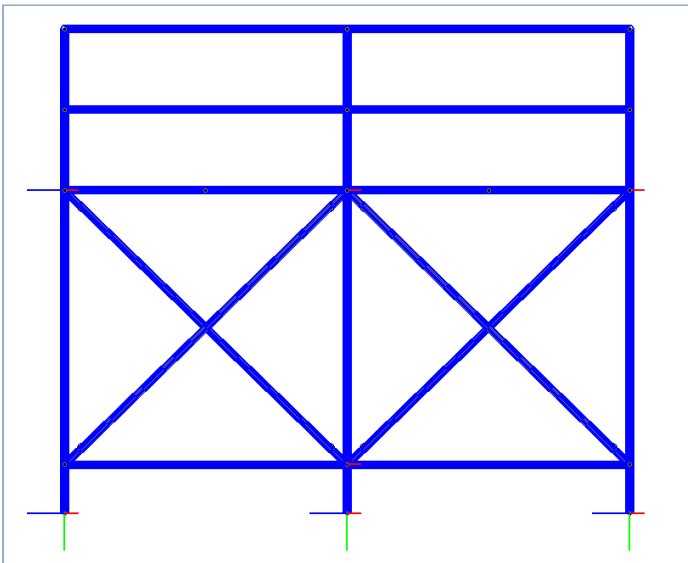
3D Isometric View



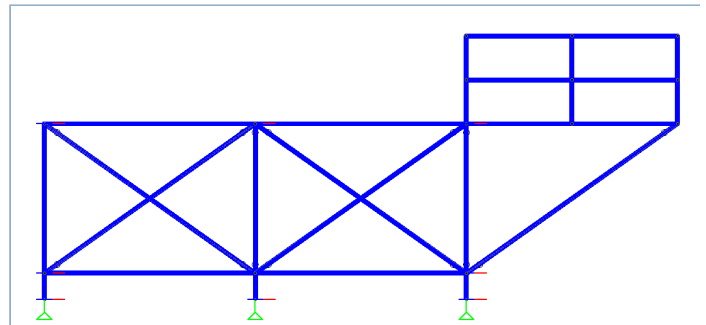
Plan View



Elevation from +X Axis



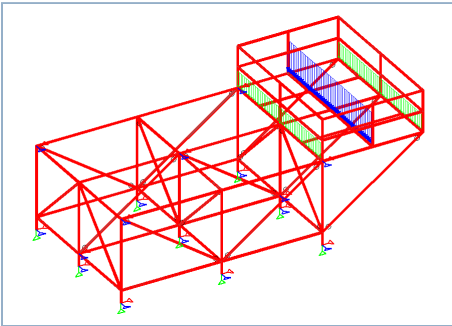
Elevation from +Z Axis



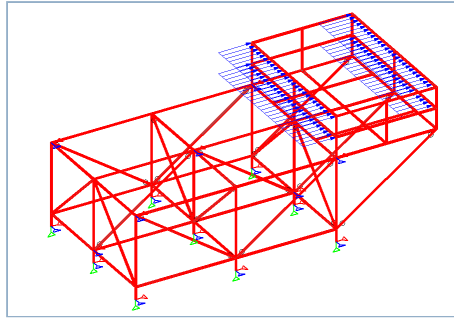
**Scaffold Dimensions:** W = 7.2m | D = 3.5m | H = 3.0m | **Lifts:** 0.3m, 2.0m, 2.5m, 3.0m

## APPENDIX B - LOAD DIAGRAMS

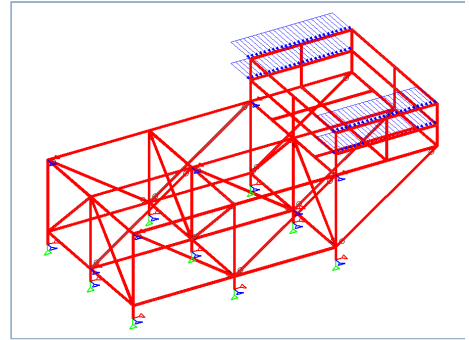
Live Load - Platform Live Load (LL)



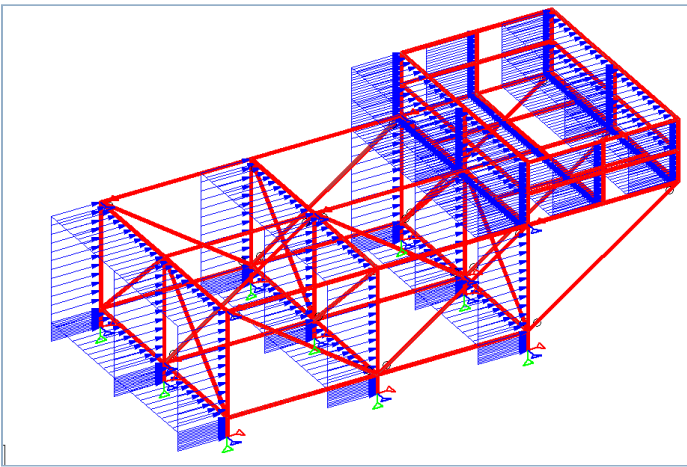
Handrail Load in X Direction (HLX)



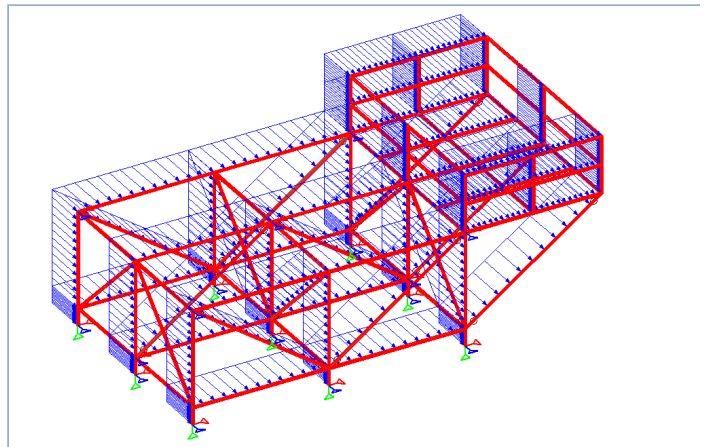
Handrail Load in Z Direction (HLZ)



Wind Load - X Axis (WLX)

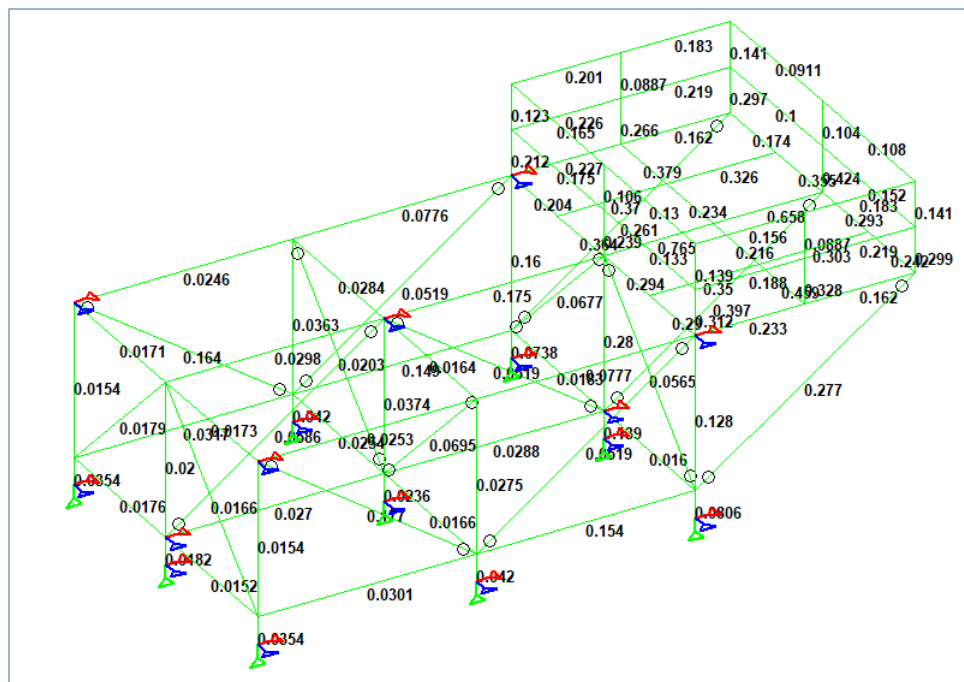


Wind Load - Z Axis (WLZ)

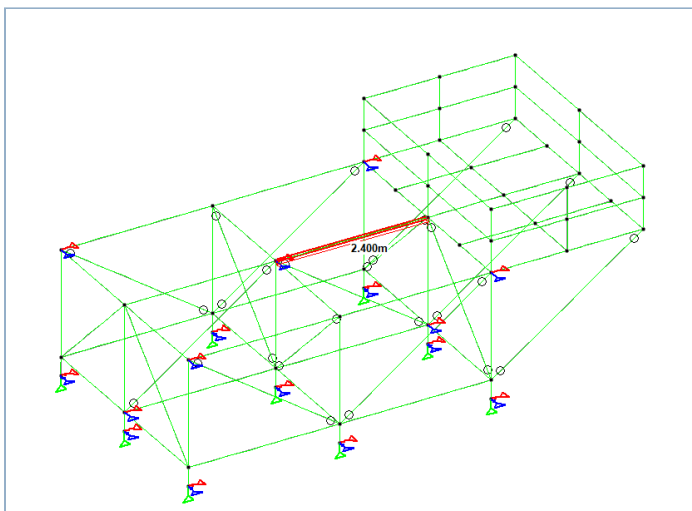


## APPENDIX C - ANALYSIS RESULTS

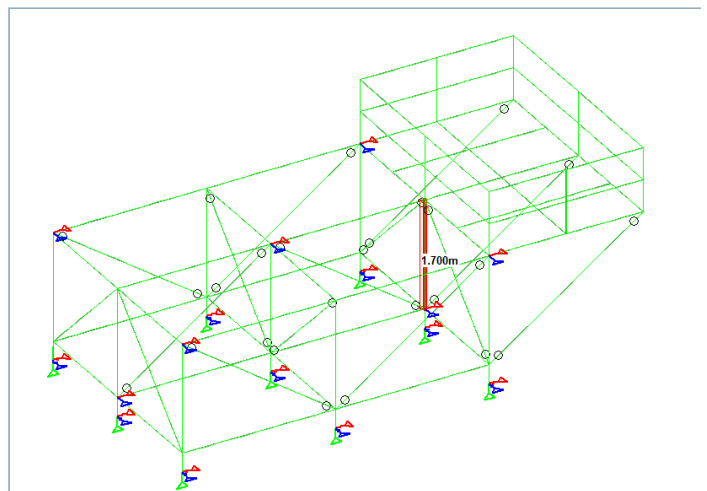
### Utilization Ratio (UC) Diagram



### Vertical Deflection Diagram



### Horizontal Deflection Diagram



## APPENDIX D - NET GLOBAL REACTION ON PARENT STRUCTURE

### Net Global Reaction on Parent Structure (X and Z)

**Definition:** Net Global Reaction = Sum FX and Sum FZ at the relevant support/tie nodes, preserving signs. Resultant H =  $\sqrt{(\text{Sum FX})^2 + (\text{Sum FZ})^2}$ . This metric is used when quoting the reaction on the parent structure because it represents the true net vector action transferred to the parent structure after directional cancellation. Relevant tie nodes: 7, 8, 19, 20, 21, 22, 27.

| Load Case               | Load Description                | Tie Support Count | Net X Reaction, Sum FX (kN) | Net Z Reaction, Sum FZ (kN) | Resultant H (kN) |
|-------------------------|---------------------------------|-------------------|-----------------------------|-----------------------------|------------------|
| LC 1                    | DL                              | 7                 | 0.01                        | -0.0                        | 0.01             |
| LC 2                    | LL                              | 7                 | -0.01                       | 0.0                         | 0.01             |
| LC 3                    | HLX                             | 7                 | -2.09                       | 0.0                         | 2.09             |
| LC 4                    | HLZ                             | 7                 | 0.0                         | -1.45                       | 1.45             |
| LC 5                    | WLX                             | 7                 | -5.37                       | -0.0                        | 5.37             |
| LC 6                    | WLZ                             | 7                 | 0.0                         | -11.64                      | 11.64            |
| LC 7                    | DL + LL                         | 7                 | 0.0                         | -0.0                        | 0.0              |
| LC 8                    | DL + HLX                        | 7                 | -2.1                        | -0.0                        | 2.1              |
| LC 9                    | DL + HLZ                        | 7                 | -0.0                        | -1.44                       | 1.44             |
| LC 10                   | DL + WLX                        | 7                 | -5.36                       | -0.0                        | 5.36             |
| LC 11                   | DL + WLZ                        | 7                 | 0.01                        | -11.63                      | 11.63            |
| LC 12                   | DL + LL + HLX                   | 7                 | -2.09                       | -0.0                        | 2.09             |
| LC 13                   | DL + LL + HLZ                   | 7                 | -0.0                        | -1.45                       | 1.45             |
| LC 14                   | DL + LL + WLX                   | 7                 | -5.37                       | -0.0                        | 5.37             |
| LC 15                   | DL + LL + WLZ                   | 7                 | 0.0                         | -11.64                      | 11.64            |
| LC 16                   | DL + LL + HLX + WLX             | 7                 | -7.47                       | -0.0                        | 7.47             |
| LC 17                   | DL + LL + HLZ + WLZ             | 7                 | -0.0                        | -13.07                      | 13.07            |
| LC 18                   | DL + LL + HLX + WLZ             | 7                 | -2.09                       | -11.63                      | 11.816           |
| LC 19                   | DL + LL + HLZ + WLX             | 7                 | -5.36                       | -1.44                       | 5.55             |
| LC 20                   | 1.5DL                           | 7                 | -0.0                        | -0.0                        | 0.0              |
| LC 21                   | 1.5DL + 1.5LL                   | 7                 | -0.02                       | -0.0                        | 0.02             |
| LC 22                   | 1.5DL + 1.5HLX                  | 7                 | -3.15                       | -0.0                        | 3.15             |
| LC 23                   | 1.5DL + 1.5HLZ                  | 7                 | 0.01                        | -2.17                       | 2.17             |
| LC 24                   | 1.5DL + 1.5WLX                  | 7                 | -8.05                       | -0.0                        | 8.05             |
| LC 25                   | 1.5DL + 1.5WLZ                  | 7                 | 0.01                        | -17.45                      | 17.45            |
| LC 26                   | 1.5DL + 1.5LL + 1.5HLX          | 7                 | -3.15                       | -0.0                        | 3.15             |
| LC 27                   | 1.5DL + 1.5LL + 1.5HLZ          | 7                 | -0.01                       | -2.16                       | 2.16             |
| LC 28                   | 1.5DL + 1.5LL + 1.5WLX          | 7                 | -8.05                       | -0.0                        | 8.05             |
| LC 29                   | 1.5DL + 1.5LL + 1.5WLZ          | 7                 | -0.01                       | -17.45                      | 17.45            |
| LC 30                   | 1.5DL + 1.5LL + 1.5HLX + 1.5WLX | 7                 | -11.2                       | -0.0                        | 11.2             |
| LC 31                   | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLZ | 7                 | -0.01                       | -19.62                      | 19.62            |
| LC 32                   | 1.5DL + 1.5LL + 1.5HLX + 1.5WLZ | 7                 | -3.16                       | -17.45                      | 17.734           |
| LC 33                   | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLX | 7                 | -8.05                       | -2.16                       | 8.335            |
| Governing Net Resultant |                                 |                   | -0.01                       | -19.62                      | 19.62            |

## APPENDIX E - TOTAL RESTRAINT DEMAND AT TIE SUPPORTS

### Total Restraint Demand at Tie Supports (X and Z)

**Definition:** Total Restraint Demand = Sum |FX| and Sum |FZ| at the relevant support/tie nodes, with no directional cancellation. Resultant H(abs) =  $\sqrt{(\text{Sum } |FX|)^2 + (\text{Sum } |FZ|)^2}$ . This metric is used separately for cumulative tie/restraint demand review because every restraint force contribution is counted by magnitude. It is not combined with, averaged with, or substituted for the Net Global Reaction on the parent structure. Relevant tie nodes: 7, 8, 19, 20, 21, 22, 27.

| Load Case                  | Load Description                | Tie Support Count | Total X(abs), Sum  FX  (kN) | Total Z(abs), Sum  FZ  (kN) | Resultant H(abs) (kN) |
|----------------------------|---------------------------------|-------------------|-----------------------------|-----------------------------|-----------------------|
| LC 1                       | DL                              | 7                 | 3.35                        | 0.4                         | 3.374                 |
| LC 2                       | LL                              | 7                 | 14.71                       | 1.32                        | 14.769                |
| LC 3                       | HLX                             | 7                 | 3.11                        | 0.38                        | 3.133                 |
| LC 4                       | HLZ                             | 7                 | 0.98                        | 1.47                        | 1.767                 |
| LC 5                       | WLX                             | 7                 | 5.37                        | 0.48                        | 5.391                 |
| LC 6                       | WLZ                             | 7                 | 3.36                        | 11.64                       | 12.115                |
| LC 7                       | DL + LL                         | 7                 | 18.06                       | 1.72                        | 18.142                |
| LC 8                       | DL + HLX                        | 7                 | 6.44                        | 0.78                        | 6.487                 |
| LC 9                       | DL + HLZ                        | 7                 | 3.34                        | 1.48                        | 3.653                 |
| LC 10                      | DL + WLX                        | 7                 | 6.8                         | 0.86                        | 6.854                 |
| LC 11                      | DL + WLZ                        | 7                 | 4.87                        | 11.63                       | 12.608                |
| LC 12                      | DL + LL + HLX                   | 7                 | 21.17                       | 2.1                         | 21.274                |
| LC 13                      | DL + LL + HLZ                   | 7                 | 18.04                       | 1.75                        | 18.125                |
| LC 14                      | DL + LL + WLX                   | 7                 | 21.47                       | 2.2                         | 21.582                |
| LC 15                      | DL + LL + WLZ                   | 7                 | 18.04                       | 11.64                       | 21.469                |
| LC 16                      | DL + LL + HLX + WLX             | 7                 | 24.57                       | 2.58                        | 24.705                |
| LC 17                      | DL + LL + HLZ + WLZ             | 7                 | 18.04                       | 13.07                       | 22.277                |
| LC 18                      | DL + LL + HLX + WLZ             | 7                 | 21.15                       | 11.63                       | 24.137                |
| LC 19                      | DL + LL + HLZ + WLX             | 7                 | 21.46                       | 2.22                        | 21.575                |
| LC 20                      | 1.5DL                           | 7                 | 5.04                        | 0.6                         | 5.076                 |
| LC 21                      | 1.5DL + 1.5LL                   | 7                 | 27.06                       | 2.6                         | 27.185                |
| LC 22                      | 1.5DL + 1.5HLX                  | 7                 | 9.67                        | 1.18                        | 9.742                 |
| LC 23                      | 1.5DL + 1.5HLZ                  | 7                 | 5.03                        | 2.23                        | 5.502                 |
| LC 24                      | 1.5DL + 1.5WLX                  | 7                 | 10.21                       | 1.3                         | 10.292                |
| LC 25                      | 1.5DL + 1.5WLZ                  | 7                 | 7.31                        | 17.45                       | 18.919                |
| LC 26                      | 1.5DL + 1.5LL + 1.5HLX          | 7                 | 31.71                       | 3.16                        | 31.867                |
| LC 27                      | 1.5DL + 1.5LL + 1.5HLZ          | 7                 | 27.07                       | 2.62                        | 27.196                |
| LC 28                      | 1.5DL + 1.5LL + 1.5WLX          | 7                 | 32.17                       | 3.28                        | 32.337                |
| LC 29                      | 1.5DL + 1.5LL + 1.5WLZ          | 7                 | 27.07                       | 17.45                       | 32.207                |
| LC 30                      | 1.5DL + 1.5LL + 1.5HLX + 1.5WLX | 7                 | 36.86                       | 3.86                        | 37.062                |
| LC 31                      | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLZ | 7                 | 27.07                       | 19.62                       | 33.432                |
| LC 32                      | 1.5DL + 1.5LL + 1.5HLX + 1.5WLZ | 7                 | 31.72                       | 17.45                       | 36.203                |
| LC 33                      | 1.5DL + 1.5LL + 1.5HLZ + 1.5WLX | 7                 | 32.17                       | 3.32                        | 32.341                |
| Governing Restraint Demand |                                 |                   | 36.86                       | 3.86                        | 37.062                |

# APPENDIX F - GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

| Beam | L/C | Node | Axial Force<br>kN | Shear-Y<br>kN | Shear-Z<br>kN | Torsion<br>kN-m | Moment-Y<br>kN-m | Moment-Z<br>kN-m |
|------|-----|------|-------------------|---------------|---------------|-----------------|------------------|------------------|
| 44   | 16  | 31   | 4.831             | -0.149        | -0.023        | -0.048          | -0.008           | 0.116            |
| 43   | 16  | 33   | 4.830             | -0.149        | 0.023         | 0.048           | 0.008            | 0.116            |
| 44   | 18  | 31   | 4.483             | -0.201        | 0.044         | -0.046          | 0.051            | 0.148            |
| 48   | 16  | 29   | 4.296             | 0.365         | -0.023        | 0.043           | -0.020           | -0.186           |
| 50   | 16  | 28   | 4.296             | 0.365         | 0.023         | -0.043          | 0.020            | -0.186           |
| 48   | 18  | 29   | 4.295             | 0.316         | -0.124        | 0.047           | -0.049           | -0.154           |
| 44   | 19  | 31   | 4.288             | -0.222        | 0.008         | -0.041          | 0.010            | 0.159            |
| 48   | 19  | 29   | 4.084             | 0.303         | -0.037        | 0.038           | -0.020           | -0.145           |
| 48   | 17  | 29   | 4.083             | 0.253         | -0.138        | 0.043           | -0.049           | -0.113           |
| 44   | 12  | 31   | 4.074             | -0.195        | -0.007        | -0.048          | -0.001           | 0.144            |
| 43   | 12  | 33   | 4.074             | -0.195        | 0.007         | 0.048           | 0.001            | 0.144            |
| 44   | 14  | 31   | 4.038             | -0.214        | -0.019        | -0.048          | -0.009           | 0.154            |
| 43   | 14  | 33   | 4.038             | -0.214        | 0.019         | 0.048           | 0.009            | 0.154            |
| 44   | 17  | 31   | 3.941             | -0.273        | 0.075         | -0.039          | 0.068            | 0.190            |
| 48   | 12  | 29   | 3.866             | 0.312         | -0.004        | 0.043           | -0.005           | -0.152           |
| 50   | 12  | 28   | 3.866             | 0.312         | 0.004         | -0.043          | 0.005            | -0.152           |
| 48   | 14  | 29   | 3.828             | 0.296         | -0.016        | 0.043           | -0.011           | -0.142           |
| 50   | 14  | 28   | 3.828             | 0.296         | 0.016         | -0.043          | 0.011            | -0.142           |
| 48   | 15  | 29   | 3.827             | 0.247         | -0.116        | 0.048           | -0.040           | -0.110           |
| 43   | 19  | 33   | 3.789             | -0.206        | 0.045         | 0.056           | 0.027            | 0.149            |
| 44   | 15  | 31   | 3.691             | -0.265        | 0.049         | -0.047          | 0.050            | 0.185            |
| 43   | 18  | 33   | 3.665             | -0.189        | 0.058         | 0.050           | 0.052            | 0.140            |
| 48   | 13  | 29   | 3.654             | 0.249         | -0.018        | 0.039           | -0.005           | -0.111           |
| 50   | 19  | 28   | 3.572             | 0.290         | -0.005        | -0.048          | 0.003            | -0.138           |
| 44   | 13  | 31   | 3.531             | -0.267        | 0.024         | -0.041          | 0.017            | 0.187            |
| 50   | 18  | 28   | 3.437             | 0.308         | -0.116        | -0.039          | -0.039           | -0.150           |
| 50   | 18  | 28   | 3.437             | 0.308         | -0.116        | -0.039          | -0.039           | -0.150           |



# Vertical Displacement Table (SLS Combinations)

| Beam | Max Disp<br>mm | Location<br>m | L/C | L/Displ | Global X<br>mm | Global Y<br>mm | Global Z<br>mm |
|------|----------------|---------------|-----|---------|----------------|----------------|----------------|
| 41   | 2.059          | 1.600         | 16  | 1165    | 0.043          | 2.016          | 0.000          |
| 41   | 1.860          | 1.600         | 18  | 1290    | 0.034          | 1.805          | 0.209          |
| 41   | 1.849          | 1.600         | 12  | 1298    | 0.034          | 1.805          | 0.000          |
| 41   | 1.712          | 1.600         | 19  | 1402    | 0.036          | 1.668          | -0.007         |
| 41   | 1.712          | 1.600         | 14  | 1402    | 0.036          | 1.668          | 0.000          |
| 41   | 1.515          | 1.600         | 15  | 1584    | 0.027          | 1.457          | 0.209          |
| 41   | 1.514          | 1.600         | 17  | 1585    | 0.027          | 1.457          | 0.203          |
| 41   | 1.501          | 1.600         | 13  | 1598    | 0.027          | 1.457          | -0.007         |
| 41   | 1.501          | 1.600         | 7   | 1598    | 0.027          | 1.457          | 0.000          |
| 49   | 1.373          | 0.400         | 17  | 873     | 0.069          | -7.620         | 0.775          |
| 49   | 1.372          | 0.400         | 15  | 874     | 0.069          | -7.620         | 0.578          |
| 49   | 1.370          | 0.400         | 13  | 875     | 0.069          | -7.620         | 0.196          |
| 49   | 1.370          | 0.400         | 7   | 875     | 0.069          | -7.620         | 0.000          |
| 49   | 1.209          | 0.400         | 19  | 992     | 0.083          | -7.389         | 0.196          |
| 49   | 1.209          | 0.400         | 14  | 992     | 0.083          | -7.389         | 0.000          |
| 45   | 1.171          | 0.800         | 16  | 1024    | 0.079          | -7.117         | 0.000          |
| 49   | 1.124          | 0.400         | 18  | 1067    | 0.080          | -7.285         | 0.578          |
| 49   | 1.122          | 0.400         | 12  | 1069    | 0.080          | -7.285         | 0.000          |
| 45   | 1.090          | 0.900         | 18  | 1100    | 0.067          | -7.858         | 0.596          |
| 45   | 1.088          | 0.900         | 12  | 1103    | 0.067          | -7.858         | 0.000          |
| 109  | 1.049          | 0.700         | 17  | 1143    | 0.049          | -6.632         | 0.679          |
| 109  | 1.037          | 0.700         | 13  | 1156    | 0.049          | -6.541         | 0.168          |
| 45   | 1.033          | 0.900         | 19  | 1161    | 0.070          | -7.822         | 0.212          |
| 45   | 1.033          | 0.900         | 14  | 1161    | 0.070          | -7.822         | 0.000          |
| 109  | 1.032          | 0.700         | 15  | 1162    | 0.050          | -6.502         | 0.507          |
| 115  | 1.021          | 0.500         | 7   | 1175    | 0.050          | -6.410         | 0.005          |

Max = 2.059 mm (LC 16) | L = Member 41 length (2400 mm) | Allowable L/100 = 24.0 mm -> **PASS**



## Horizontal Displacement Table (SLS Combinations)

| Beam | Max Disp<br>mm | Location<br>m | L/C | L/Displ | Global X<br>mm | Global Y<br>mm | Global Z<br>mm |
|------|----------------|---------------|-----|---------|----------------|----------------|----------------|
| 29   | 1.554          | 0.992         | 16  | 1093    | -1.517         | -0.044         | 0.000          |
| 29   | 1.500          | 0.992         | 18  | 1133    | -1.470         | -0.044         | -0.025         |
| 29   | 1.499          | 0.992         | 12  | 1133    | -1.470         | -0.044         | 0.000          |
| 29   | 1.296          | 1.133         | 19  | 1311    | -1.259         | -0.048         | -0.063         |
| 29   | 1.294          | 1.133         | 14  | 1313    | -1.259         | -0.048         | 0.000          |
| 29   | 1.243          | 0.992         | 17  | 1368    | -1.216         | -0.044         | -0.086         |
| 29   | 1.241          | 0.992         | 13  | 1370    | -1.216         | -0.044         | -0.061         |
| 29   | 1.240          | 0.992         | 15  | 1371    | -1.216         | -0.044         | -0.025         |
| 29   | 1.239          | 0.992         | 7   | 1371    | -1.216         | -0.044         | 0.000          |
| 23   | 0.618          | 1.133         | 17  | 2752    | -0.318         | -0.043         | -0.554         |
| 23   | 0.574          | 1.133         | 18  | 2959    | -0.427         | -0.046         | -0.436         |
| 23   | 0.543          | 0.992         | 13  | 3129    | -0.360         | -0.044         | -0.458         |
| 23   | 0.529          | 1.133         | 19  | 3214    | -0.317         | -0.045         | -0.459         |
| 24   | 0.523          | 0.992         | 12  | 3248    | -0.474         | -0.046         | 0.340          |
| 23   | 0.523          | 0.992         | 12  | 3249    | -0.474         | -0.046         | -0.340         |
| 23   | 0.519          | 1.133         | 15  | 3274    | -0.325         | -0.047         | -0.439         |
| 24   | 0.502          | 0.992         | 16  | 3386    | -0.446         | -0.044         | 0.338          |
| 23   | 0.502          | 0.992         | 16  | 3387    | -0.446         | -0.044         | -0.338         |
| 24   | 0.481          | 0.992         | 18  | 3530    | -0.488         | -0.051         | 0.263          |
| 24   | 0.454          | 0.992         | 7   | 3740    | -0.369         | -0.047         | 0.343          |
| 23   | 0.454          | 0.992         | 7   | 3740    | -0.369         | -0.047         | -0.343         |
| 24   | 0.437          | 1.133         | 14  | 3892    | -0.325         | -0.048         | 0.344          |
| 23   | 0.437          | 1.133         | 14  | 3893    | -0.325         | -0.048         | -0.344         |
| 63   | 0.433          | 0.208         | 16  | 1154    | -0.249         | -0.503         | 0.000          |
| 24   | 0.403          | 0.992         | 15  | 4215    | -0.383         | -0.051         | 0.267          |
| 63   | 0.388          | 0.208         | 18  | 1287    | -0.554         | -0.479         | 0.340          |

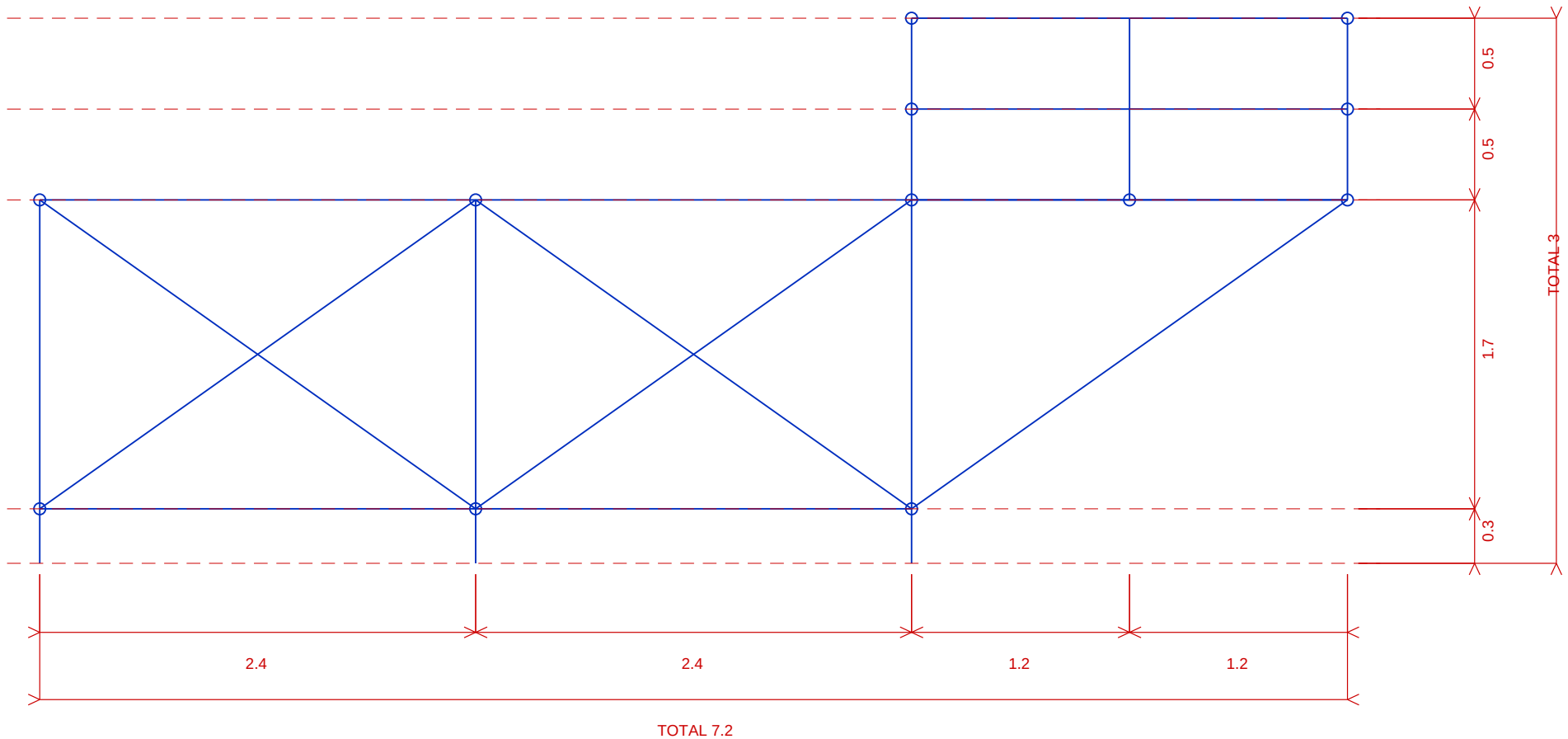
Max = **1.534 mm** (LC 16) | L = Member 29 length (1700 mm) | Allowable L/200 = 8.5 mm -> **PASS**

## APPENDIX G - REFERENCES & STANDARDS

| Reference           | Title                                                                               | Year |
|---------------------|-------------------------------------------------------------------------------------|------|
| BS EN 12811-1:2003  | Temporary Works Equipment - Scaffolds - Performance Requirements and General Design | 2003 |
| BS EN 1993-1-1:2005 | Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings      | 2005 |
| BS EN 1991-1-4:2005 | Eurocode 1: Actions on Structures - Wind Actions                                    | 2005 |
| BS EN 1990:2002     | Eurocode: Basis of Structural Design                                                | 2002 |
| NASC TG20:13        | Guide to Good Practice for Scaffolding with Tubes and Fittings                      | 2013 |
| Dredger P.          | How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust                     | -    |
| STAAD.Pro CONNECT   | STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software             | 2022 |

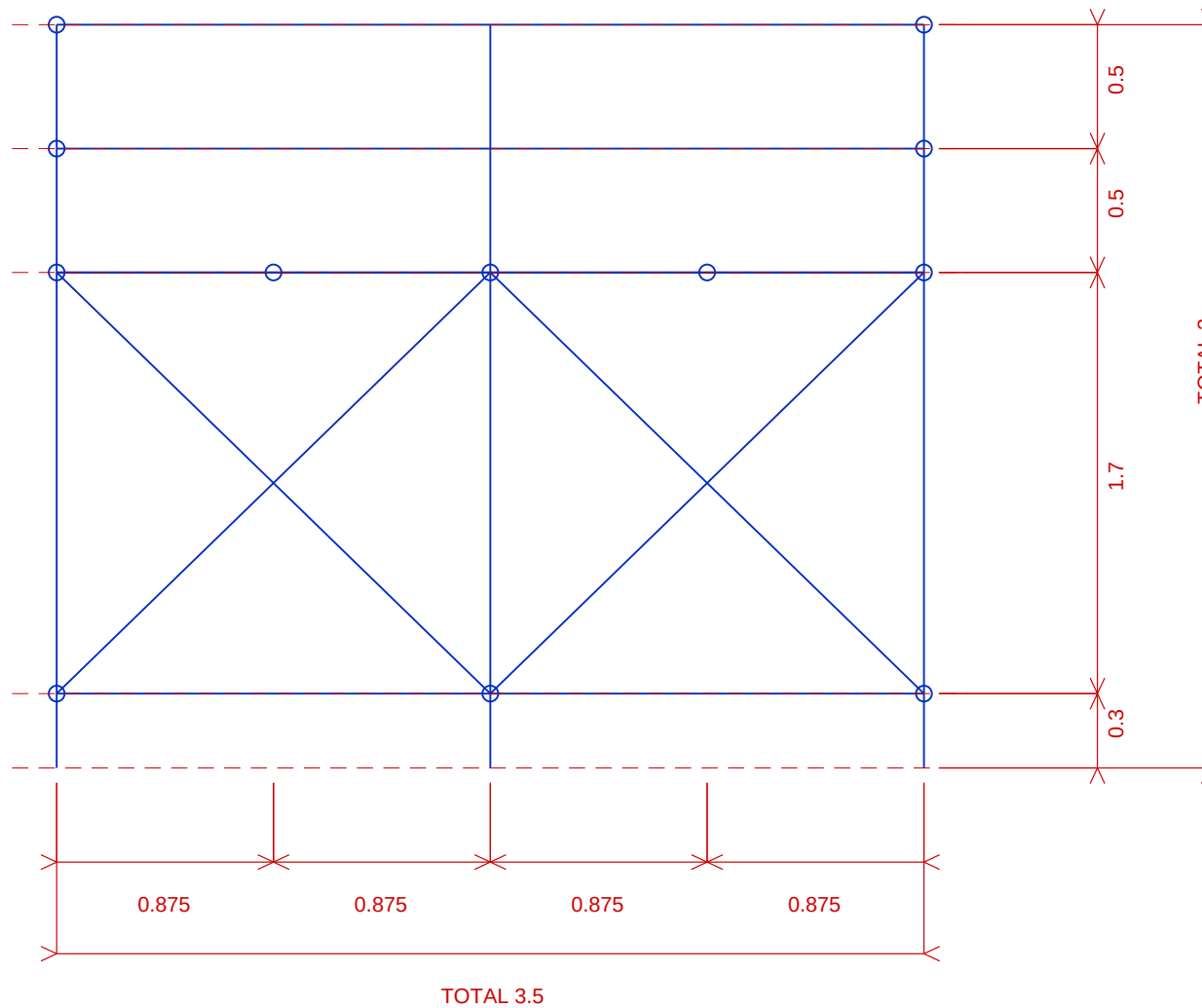
**Software Used:** STAAD.Pro CONNECT Edition V22 (Bentley Systems) - 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

**Report:** Scaffold Structural Design Report - ARCO M&E, NLNG Bonny Island.



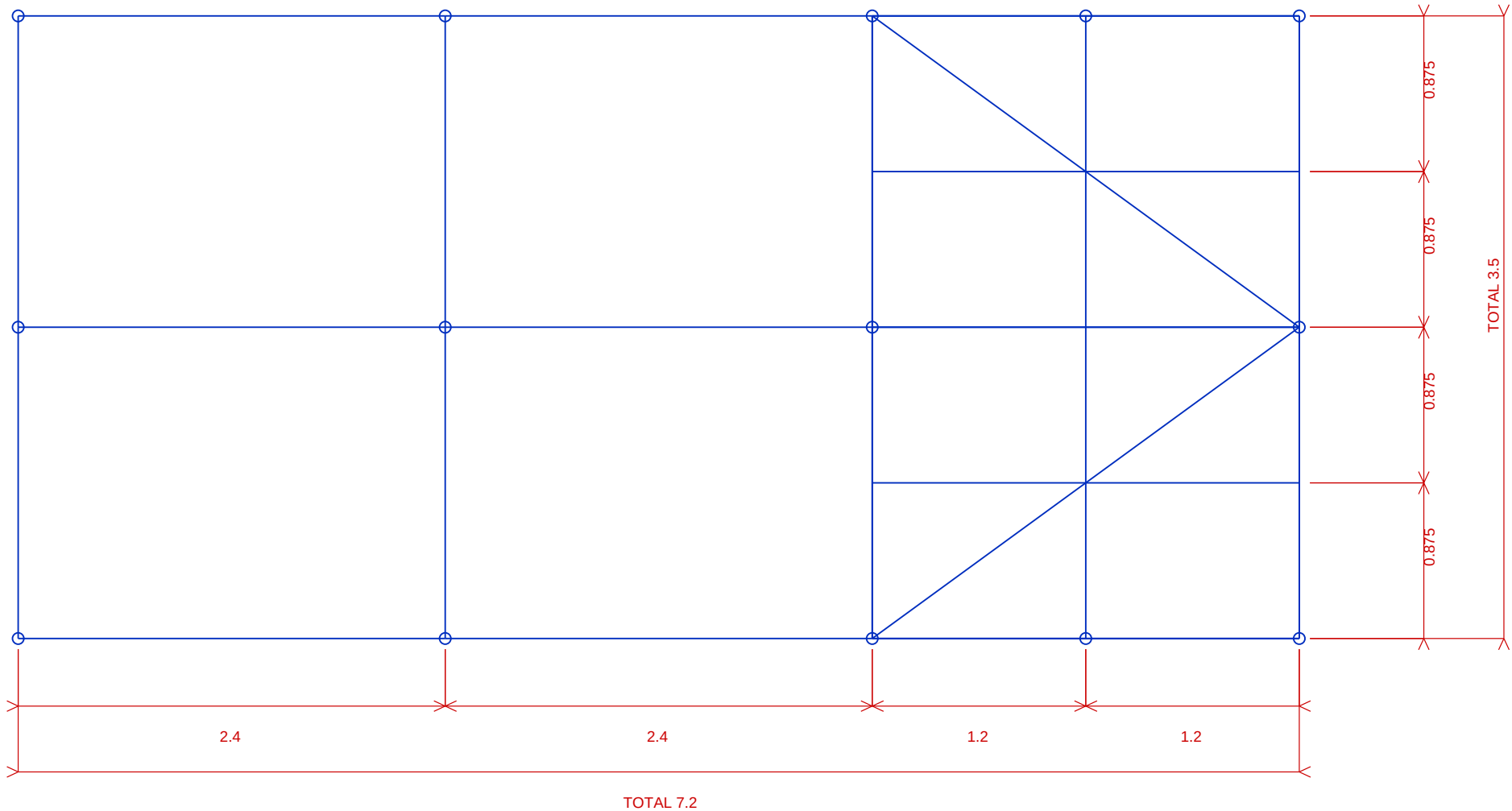
FRONT VIEW

|                                                                   |  |                                                  |                                  |                 |                                                                                                   |                                                                                       |                   |
|-------------------------------------------------------------------|--|--------------------------------------------------|----------------------------------|-----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|
| PROJECT<br><b>SCAFFOLDING WORKS</b>                               |  | DRAWING NO.<br><b>PMS-WA-ARCO-003-FRONT-VIEW</b> |                                  |                 |  <b>NLNG</b> |  |                   |
| DRAWING TITLE<br><b>SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW</b> |  | SCALE<br><b>NTS</b>                              | DATE<br><b>19-JUN-2026</b>       | REV<br><b>0</b> |                                                                                                   |                                                                                       |                   |
| CLIENT                                                            |  | LOCATION                                         | DRAWN<br><b>Nnamani, Ifeanyi</b> | CHECKED         | APPROVED                                                                                          | COMPANY                                                                               |                   |
|                                                                   |  |                                                  |                                  |                 |                                                                                                   | SHEET<br><b>1 OF 4</b>                                                                | SIZE<br><b>A4</b> |



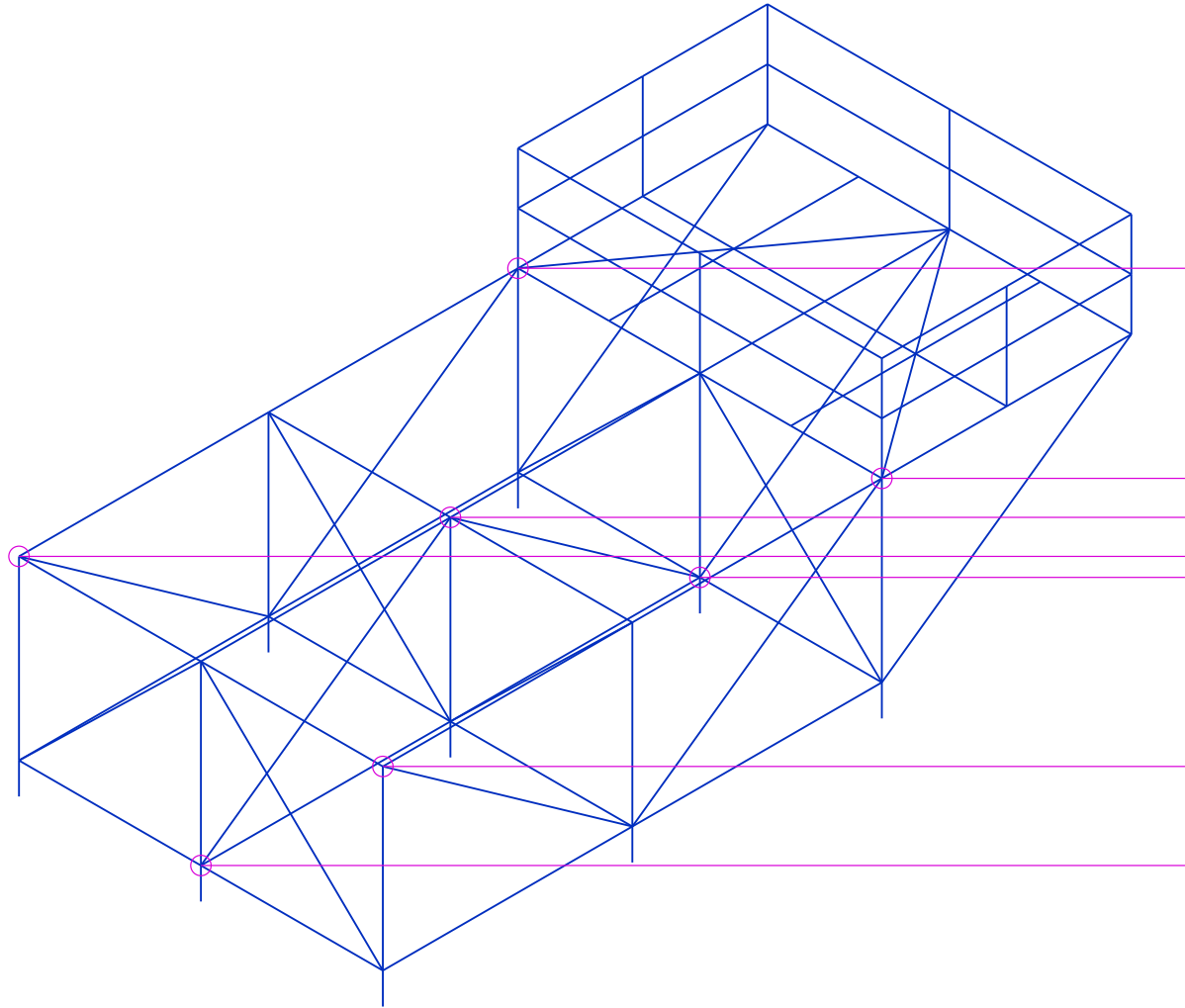
SIDE VIEW

|                                          |          |                           |             |          |                                                                                       |                                                                                       |
|------------------------------------------|----------|---------------------------|-------------|----------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| PROJECT                                  |          | DRAWING NO.               |             |          |  |  |
| SCAFFOLDING WORKS                        |          | PMS-WA-ARCO-003-SIDE-VIEW |             |          |                                                                                       |                                                                                       |
| DRAWING TITLE                            |          | SCALE                     | DATE        | REV      | COMPANY                                                                               |                                                                                       |
| SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW |          | NTS                       | 19-JUN-2026 | 0        |                                                                                       |                                                                                       |
| CLIENT                                   | LOCATION | DRAWN                     | CHECKED     | APPROVED | SHEET                                                                                 |                                                                                       |
|                                          |          | Nnamani, Ifeanyi          |             |          |                                                                                       |                                                                                       |
|                                          |          |                           |             |          | 2 OF 4                                                                                | A4                                                                                    |



PLAN VIEW

|                                          |          |                           |             |          |                                                                                       |                                                                                       |
|------------------------------------------|----------|---------------------------|-------------|----------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| PROJECT                                  |          | DRAWING NO.               |             |          |  |  |
| SCAFFOLDING WORKS                        |          | PMS-WA-ARCO-003-PLAN-VIEW |             |          |                                                                                       |                                                                                       |
| DRAWING TITLE                            |          | SCALE                     | DATE        | REV      | COMPANY                                                                               |                                                                                       |
| SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW |          | NTS                       | 19-JUN-2026 | 0        |                                                                                       |                                                                                       |
| CLIENT                                   | LOCATION | DRAWN                     | CHECKED     | APPROVED | SHEET                                                                                 | SIZE                                                                                  |
|                                          |          | Nnamani, Ifeanyi          |             |          | 3 OF 4                                                                                | A4                                                                                    |



3D VIEW

|                                                                |  |                                               |                                  |                 |                                                                                                   |                                                                                       |                   |
|----------------------------------------------------------------|--|-----------------------------------------------|----------------------------------|-----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|
| PROJECT<br><b>SCAFFOLDING WORKS</b>                            |  | DRAWING NO.<br><b>PMS-WA-ARCO-003-3D-VIEW</b> |                                  |                 |  <b>NLNG</b> |  |                   |
| DRAWING TITLE<br><b>SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW</b> |  | SCALE<br><b>NTS</b>                           | DATE<br><b>19-JUN-2026</b>       | REV<br><b>0</b> |                                                                                                   |                                                                                       |                   |
| CLIENT                                                         |  | LOCATION                                      | DRAWN<br><b>Nnamani, Ifeanyi</b> | CHECKED         | APPROVED                                                                                          | COMPANY                                                                               |                   |
|                                                                |  |                                               |                                  |                 |                                                                                                   | SHEET<br><b>4 OF 4</b>                                                                | SIZE<br><b>A4</b> |

## 3D Isometric View

